

Paper F-03: Black Hole Thermodynamics: A Constraint-Driven Flux Dynamics (CDFS) Perspective

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper asks whether Constraint-Driven Flux Dynamics (CDFS) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux Φ , constraint C , surface responsiveness S , and structural memory M_s . The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail?

1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

Notation. In Part IV, C names the active constraint or capacity burden in the system being discussed. Φ is the driving flow, S is the surface response, and M_s is retained structural memory. Λ appears only where the Part II Life Number is being referenced [2]. Other Greek symbols are local coefficients whose meaning is set by the equation in which they appear.

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \rightarrow \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

In standard relativistic frameworks, a black hole is a region of spacetime exhibiting such strong gravitational acceleration that nothing, including light, can escape. The boundary defining this region is the event horizon. Following the works of Bekenstein and Hawking, the event horizon is understood to possess an entropy proportional to its surface area, $S_{BH} = \frac{k_B A}{4l_p^2}$, suggesting that all the information of the infalling volume is holographically encoded on the 2D boundary.

The CDFD paradigm posits that the universe is governed by the continuous interaction of flow (Φ) and constraint (C) across adaptive surfaces. In this context, a black hole is not a singularity of broken physics; rather, it is the maximal boundary condition of the CDFD equations. It is the extreme limit where constraint C approaches infinity for classical flow, forcing all throughput to undergo quantum phase transitions.

3 The Event Horizon as an Active Surface

In the CDFD framework, any boundary separating two topological regions evaluates to an active surface governed by the equation:

$$\Psi_s = \left(\frac{\Phi}{C} \right) \cdot S \cdot M_s \quad (1)$$

For an event horizon, the parameters map as follows:

- **Flux (Φ):** The accretion of matter and information crossing the boundary (inward), and Hawking radiation (outward).
- **Constraint (C):** The causal barrier generated by extreme spacetime curvature. For macroscopic inward flow, the boundary allows absolute transmission. For outward classical flow, $C \rightarrow \infty$.
- **Responsiveness (S):** The geometric surface area of the horizon A , which defines the thermodynamic capacity to process and encode incoming flux.
- **Memory (M_s):** The holographic encoding. Every bit of information that crosses the horizon is chronicled on the surface. Therefore, the event horizon possesses perfect, unbroken memory, $M_s \rightarrow 1.0$.

3.1 Information Bottlenecks and The Bekenstein Bound

As matter falls into the black hole, the inward flux Φ_{in} causes the mass, and thus the surface area S , to increase. This constitutes a dynamic adaptation. The horizon expands its surface

responsiveness precisely to accommodate the increased memory load M_s . The Bekenstein bound is essentially the limit of information density allowed by the constraint C .

If the black hole attempts to absorb flux faster than the surface can expand and encode it, the system would theoretically experience a $\Psi_s > 1$ "over-flux" anomaly. However, the geometric rigidity of General Relativity ensures that S expands in lockstep with mass, maintaining $\Psi_s \approx 1$.

4 Hawking Radiation as Regime Stabilization

If an isolated black hole ceases to accrete matter ($\Phi_{in} = 0$), the static nature of the system threatens to drop the adaptive ratio $\Psi_s \ll 1$. In biological systems (AFL), a drop in throughput leads to decay. In the cosmological vacuum, the event horizon undergoes analogous decay to shed its memory load and lower its constraint profile.

Hawking radiation (Φ_{out}) emerges as the necessary quantum leakage required to satisfy the CDFD equation. To prevent $\Psi_s \rightarrow 0$, the vacuum surface generates an outward flux at the expense of its own structural memory (mass/area). As the horizon shrinks, the temperature increases, amplifying Φ_{out} until complete evaporation occurs.

5 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

5.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub $\Psi_s = 14.8$, peak network $\Psi_s = 14.8$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

6 Conclusion

Black hole thermodynamics is entirely isomorphic to the constraint-flow dynamics observed in living biological systems. An event horizon is an active biological-like membrane for the cosmos: it accretes information, encodes it into structural memory, and actively radiates energy to prevent catastrophic thermodynamic stagnation. The holographic principle is merely the memory term M_s operating at the Planck limit.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs/`, and `Part_E_Synthesis/figures/`.

7 Reference Layer

References

- [1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.
- [2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.
- [3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.
- [4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.